

Lines & Angles — Full Solutions & Explanations

Digital SAT Style | Complete Working for Every Question

Q1 — EASY

■ **Correct Answer: B) 18**

STEP-BY-STEP SOLUTION:

1. Vertically opposite angles are EQUAL.
2. $3x + 12 = 5x - 24 \rightarrow 36 = 2x \rightarrow x = 18$
3. Check: $3(18)+12 = 66^\circ$ and $5(18)-24 = 66^\circ$ ✓

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $x=15 \rightarrow 57^\circ$ vs 51° , not equal.
- ✗ C: $x=20 \rightarrow 72^\circ$ vs 76° , not equal.
- ✗ D: $x=24 \rightarrow 84^\circ$ vs 96° , not equal.

■ **TRAP ALERT:** Students set angles as supplementary (sum= 180°) instead of equal.

■ **KEY INSIGHT:** Vertical angles are ALWAYS equal. Adjacent angles at an intersection are supplementary.

■ **FAST METHOD:** Vertical $\rightarrow$ equal $\rightarrow$ set expressions equal $\rightarrow$ solve.

Q2 — EASY

■ **Correct Answer: C) 212°**

STEP-BY-STEP SOLUTION:

1. Two lines create 2 pairs of vertical angles: $74^\circ, 74^\circ$ and $106^\circ, 106^\circ$.
2. The two non- 74° angles each = $180^\circ - 74^\circ = 106^\circ$.
3. Sum = $106^\circ + 106^\circ = 212^\circ$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: 74° is just one angle, not the sum.
- ✗ B: 106° is ONE remaining angle, not both.
- ✗ D: $148^\circ = 2 \times 74^\circ$, incorrectly doubles the given angle.

■ **TRAP ALERT:** DSAT asks for the SUM of two angles, not a single angle. Read carefully.

■ **KEY INSIGHT:** At any intersection: total = 360° . Two angles are θ , two are $(180^\circ - \theta)$.

■ **FAST METHOD:** $360^\circ - 74^\circ - 74^\circ = 212^\circ$.

Q3 — EASY

■ **Correct Answer: C) 25**

STEP-BY-STEP SOLUTION:

1. Co-interior angles sum to 180° .
2. $(4k-10) + (2k+40) = 180 \rightarrow 6k + 30 = 180 \rightarrow k = 25$
3. Check: $90^\circ + 90^\circ = 180^\circ$ ✓

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $k=20 \rightarrow 70^\circ + 80^\circ = 150^\circ \neq 180^\circ$.

✗ B: $k=22 \rightarrow 78^\circ+84^\circ = 162^\circ \neq 180^\circ$.

✗ D: $k=30 \rightarrow 110^\circ+100^\circ = 210^\circ \neq 180^\circ$.

■ **TRAP ALERT:** Setting co-interior angles EQUAL (like alternate interior) gives $k \approx 16.7$.

■ **KEY INSIGHT:** "C" shape = co-interior = supplementary (180°). "Z" shape = alternate interior = equal.

■ **FAST METHOD:** Co-interior = sum to 180. One equation, one unknown.

Q4 — EASY

■ **Correct Answer: B) 40°**

STEP-BY-STEP SOLUTION:

1. Let angles be $2x$ and $7x$. Sum = 180° .
2. $9x = 180 \rightarrow x = 20$. Smaller = $2(20) = 40^\circ$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: 20° is the value of x , not $2x$.
- ✗ C: $60^\circ+120^\circ=180^\circ$ but ratio is 1:2, not 2:7.
- ✗ D: 140° is the LARGER angle (7×20).

■ **TRAP ALERT:** Answering $x=20$ instead of $2x=40$.

■ **KEY INSIGHT:** Supplementary= 180° . Complementary= 90° . In ratio problems, assign parts then multiply.

■ **FAST METHOD:** Smaller = $(2/9) \times 180^\circ = 40^\circ$.

Q5 — EASY

■ **Correct Answer: C) 148°**

STEP-BY-STEP SOLUTION:

1. BD bisects ABE $\rightarrow$ angle ABD = $64^\circ/2 = 32^\circ$.
2. A, B, C collinear $\rightarrow$ straight line = 180° .
3. Angle DBC = $180^\circ - 32^\circ = 148^\circ$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: 32° is the bisected half-angle, not DBC.
- ✗ B: $116^\circ = 180^\circ - 64^\circ =$ angle EBC, not DBC.
- ✗ D: 158° is an arithmetic error.

■ **TRAP ALERT:** Computing 32° and stopping, not realizing DBC = $180^\circ - 32^\circ$.

■ **KEY INSIGHT:** Find the half-angle from the bisector, then use the straight line (180°) to find the other side.

■ **FAST METHOD:** $180^\circ - (64^\circ/2) = 180^\circ - 32^\circ = 148^\circ$.

Q6 — MEDIUM

■ **Correct Answer: B) 54°**

STEP-BY-STEP SOLUTION:

1. Alternate interior angles are EQUAL.
2. $7m - 9 = 4m + 18 \rightarrow 3m = 27 \rightarrow m = 9$
3. Angle = $7(9) - 9 = 54^\circ$. Verify: $4(9)+18 = 54^\circ$ ✓

WHY EACH WRONG ANSWER IS WRONG:

✗ A: 49° from arithmetic errors.

✗ C: 58° would need $m \approx 9.6$.

✗ D: $63^\circ = 7(9)$, forgetting to subtract 9.

■ **TRAP ALERT:** Selecting $m=9$ or $7 \times 9=63$ instead of fully computing $7(9)-9=54$.

■ **KEY INSIGHT:** Always substitute back into BOTH expressions to verify equality.

■ **FAST METHOD:** Alternate interior $\rightarrow$ equal $\rightarrow$ set equal $\rightarrow$ solve $\rightarrow$ plug back.

Q7 — MEDIUM

■ **Correct Answer: C) Corresponding; equal.**

STEP-BY-STEP SOLUTION:

1. Angles in the same position at each intersection = corresponding angles.
2. Corresponding angles between parallel lines are EQUAL.
3. $6x+5 = 3x+20 \rightarrow x=5$; each angle = 35° .

WHY EACH WRONG ANSWER IS WRONG:

✗ A: Alternate interior are on opposite sides between the lines.

✗ B: Co-interior are between the lines on the same side.

✗ D: Complementary is not a parallel-line angle relationship.

■ **TRAP ALERT:** Confusing angle types by position. "Same position" = corresponding.

■ **KEY INSIGHT:** Four types: Corresponding (equal), Alternate interior (equal), Co-interior (supplementary), Vertical (equal).

■ **FAST METHOD:** Same position at each intersection $\rightarrow$ corresponding $\rightarrow$ equal.

Q8 — MEDIUM

■ **Correct Answer: B) 10**

STEP-BY-STEP SOLUTION:

1. Alternate exterior angles are EQUAL.
2. $9a - 7 = 5a + 33 \rightarrow 4a = 40 \rightarrow a = 10$
3. Check: $9(10)-7 = 83^\circ$ and $5(10)+33 = 83^\circ$ ✓

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $a=8 \rightarrow 65^\circ$ vs 73° .

✗ C: $a=12 \rightarrow 101^\circ$ vs 93° .

✗ D: $a=14 \rightarrow 119^\circ$ vs 103° .

■ **TRAP ALERT:** Confusing alternate exterior (equal) with same-side exterior (supplementary).

■ **KEY INSIGHT:** "Alternate" = EQUAL. "Same-side/Co-" = supplementary.

■ **FAST METHOD:** Alternate $\rightarrow$ equal $\rightarrow$ set equal $\rightarrow$ solve.

Q9 — MEDIUM

■ **Correct Answer: B) 19**

STEP-BY-STEP SOLUTION:

1. Complementary angles sum to 90° .

2. $(2y+15) + (3y-20) = 90 \rightarrow 5y - 5 = 90 \rightarrow y = 19$

3. Check: $53^\circ + 37^\circ = 90^\circ \checkmark$

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $y=15 \rightarrow 45^\circ+25^\circ=70^\circ \neq 90^\circ$.

✗ C: $y=21 \rightarrow 57^\circ+43^\circ=100^\circ \neq 90^\circ$.

✗ D: $y=24 \rightarrow 63^\circ+52^\circ=115^\circ \neq 90^\circ$.

■ **TRAP ALERT:** Using 180° (supplementary) instead of 90° (complementary).

■ **KEY INSIGHT:** COMPLEMENTARY= 90° (C before S, $90 < 180$). SUPPLEMENTARY= 180° .

■ **FAST METHOD:** Complement $\rightarrow$ sum= $90 \rightarrow$ solve.

Q10 — MEDIUM

■ **Correct Answer: C) 40**

STEP-BY-STEP SOLUTION:

1. Angles on one side of a straight line sum to 180° .

2. $(2x+10)+(x+20)+(x-10) = 180 \rightarrow 4x+20 = 180 \rightarrow x = 40$

3. Check: $90^\circ+60^\circ+30^\circ = 180^\circ \checkmark$

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $x=30 \rightarrow 70^\circ+50^\circ+20^\circ=140^\circ$.

✗ B: $x=35 \rightarrow 80^\circ+55^\circ+25^\circ=160^\circ$.

✗ D: $x=45 \rightarrow 100^\circ+65^\circ+35^\circ=200^\circ$.

■ **TRAP ALERT:** Using 360° (full rotation) instead of 180° (one side of a straight line).

■ **KEY INSIGHT:** One side of a straight line = 180° . Around a point = 360° .

■ **FAST METHOD:** Sum = $180 \rightarrow$ solve.

Q11 — MEDIUM

■ **Correct Answer: B) 88°**

STEP-BY-STEP SOLUTION:

1. Alternate interior $\rightarrow$ EQUAL.

2. $5t+8 = 3t+40 \rightarrow 2t = 32 \rightarrow t = 16$

3. Angle AEF = $5(16)+8 = 88^\circ$. Verify: $3(16)+40 = 88^\circ \checkmark$

WHY EACH WRONG ANSWER IS WRONG:

✗ A: 82° needs $t \approx 14.8$.

✗ C: 93° needs $t=17$.

✗ D: 98° needs $t=18$.

■ **TRAP ALERT:** Reporting $t=16$ instead of the ANGLE= 88° .

■ **KEY INSIGHT:** Solve for variable, then SUBSTITUTE BACK to find the angle.

■ **FAST METHOD:** $5t+8=3t+40 \rightarrow t=16 \rightarrow$ angle= 88° .

Q12 — MEDIUM

■ **Correct Answer: A) 19°**

STEP-BY-STEP SOLUTION:

1. Let s = smaller. Larger = $3s+24$.
2. $s + (3s+24) = 100 \rightarrow 4s = 76 \rightarrow s = 19^\circ$
3. Check: $19^\circ + 81^\circ = 100^\circ \checkmark$

WHY EACH WRONG ANSWER IS WRONG:

✗ B: $22^\circ \rightarrow$ larger= 90° , total= 112° .

✗ C: $25^\circ \rightarrow$ larger= 99° , total= 124° .

✗ D: $28^\circ \rightarrow$ larger= 108° , total= 136° .

■ **TRAP ALERT:** Using 180° instead of the given 100° .

■ **KEY INSIGHT:** Use the GIVEN total angle, not 180° or 90° .

■ **FAST METHOD:** $s+(3s+24)=100 \rightarrow s=19^\circ$.

Q13 — MEDIUM

■ **Correct Answer: C) 125°**

STEP-BY-STEP SOLUTION:

1. Co-interior angles are supplementary.
2. Co-interior angle = $180^\circ - 55^\circ = 125^\circ$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: 55° is the corresponding or alternate angle, not co-interior.

✗ B: $115^\circ = 180^\circ - 65^\circ$, wrong base.

✗ D: $135^\circ = 180^\circ - 45^\circ$, wrong base.

■ **TRAP ALERT:** Thinking "same-side" means equal. Co-interior = supplementary.

■ **KEY INSIGHT:** Co-interior (same-side interior) always sum to 180° .

■ **FAST METHOD:** $180^\circ - 55^\circ = 125^\circ$.

Q14 — MEDIUM

■ **Correct Answer: B) 28°**

STEP-BY-STEP SOLUTION:

1. P,Q,R collinear $\rightarrow PQS + SQR = 180^\circ$.
2. $(6n+4)+(3n-4) = 180 \rightarrow 9n = 180 \rightarrow n = 20$
3. $SQR = 3(20)-4 = 56^\circ$. QT bisects $\rightarrow SQT = 28^\circ$.

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $24^\circ = n+4$, miscalculation.

✗ C: 32° confuses with half of ABE-type.

✗ D: 36° has no basis.

■ **TRAP ALERT:** Finding $n=20$ and reporting it instead of computing the angle and halving.

■ **KEY INSIGHT:** Bisect = divide by 2. Find the angle first, then halve.

■ **FAST METHOD:** Straight line $\rightarrow n=20 \rightarrow SQR=56^\circ \rightarrow$ half= 28° .

Q15 — HARD

■ **Correct Answer: C) 27.5**

STEP-BY-STEP SOLUTION:

1. Co-interior sum to 180° .
2. $63 + (4p+7) = 180 \rightarrow 4p+70 = 180 \rightarrow p = 27.5$
3. Check: $63^\circ + 117^\circ = 180^\circ \checkmark$

WHY EACH WRONG ANSWER IS WRONG:

✗ A: $p=23.5 \rightarrow \text{sum}=164^\circ$.

✗ B: $p=25 \rightarrow \text{sum}=170^\circ$.

✗ D: $p=30 \rightarrow \text{sum}=190^\circ$.

■ **TRAP ALERT:** Setting co-interior EQUAL: $63=4p+7 \rightarrow p=14$.

■ **KEY INSIGHT:** Co-interior = supplementary (180°).

■ **FAST METHOD:** $63+4p+7=180 \rightarrow p=27.5$.

Q16 — HARD

■ **Correct Answer:** C) 65°

STEP-BY-STEP SOLUTION:

1. Alternate exterior $\rightarrow$ EQUAL.
2. $5x+15 = 9x-25 \rightarrow 40 = 4x \rightarrow x = 10$
3. Angle = $5(10)+15 = 65^\circ$. Verify: $9(10)-25 = 65^\circ \checkmark$

WHY EACH WRONG ANSWER IS WRONG:

✗ A: 55° needs $x=8$.

✗ B: 60° needs $x=9$.

✗ D: 70° needs $x=11$.

■ **TRAP ALERT:** Using co-interior (supplementary) instead of alternate exterior (equal).

■ **KEY INSIGHT:** "Alternate" always means EQUAL.

■ **FAST METHOD:** Set equal $\rightarrow$ solve $\rightarrow$ substitute.

Q17 — HARD

■ **Correct Answer:** B) 124°

STEP-BY-STEP SOLUTION:

1. Sum of interior angles = 180° : $6x+12 = 180 \rightarrow x = 28$.
2. Interior at $(2x)^\circ$ vertex = 56° .
3. Exterior = $180^\circ - 56^\circ = 124^\circ$.
4. Or: exterior = $(3x-10)+(x+22) = 74^\circ + 50^\circ = 124^\circ \checkmark$

WHY EACH WRONG ANSWER IS WRONG:

✗ A: 118° uses wrong x .

✗ C: $128^\circ = 180^\circ - 52^\circ$, arithmetic slip.

✗ D: 136° uses wrong vertex.

■ **TRAP ALERT:** Reporting the interior angle (56°) instead of the exterior.

■ **KEY INSIGHT:** Exterior Angle Theorem: exterior = sum of two non-adjacent interior angles.

■ **FAST METHOD:** $x=28 \rightarrow \text{exterior} = 74^\circ + 50^\circ = 124^\circ$.

Q18 — HARD

■ **Correct Answer: C) 20**

STEP-BY-STEP SOLUTION:

1. Co-interior sum to 180° .
2. $(5q-10)+(3q+30) = 180 \rightarrow 8q+20 = 180 \rightarrow q = 20$
3. Check: $90^\circ+90^\circ = 180^\circ \checkmark$

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $q=16 \rightarrow 70^\circ+78^\circ=148^\circ$.
- ✗ B: $q=18 \rightarrow 80^\circ+84^\circ=164^\circ$.
- ✗ D: $q=24 \rightarrow 110^\circ+102^\circ=212^\circ$.

■ **TRAP ALERT:** Setting co-interior EQUAL instead of summing to 180° .

■ **KEY INSIGHT:** Co-interior = supplementary = sum to 180° .

■ **FAST METHOD:** $8q+20=180 \rightarrow q=20$.

Q19 — HARD

■ **Correct Answer: C) 69°**

STEP-BY-STEP SOLUTION:

1. First angle = $5(3)+3(9) = 15+27 = 42^\circ$.
2. Second angle = $180^\circ-42^\circ = 138^\circ$.
3. Each half = $138^\circ/2 = 69^\circ$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: 57° from wrong x value.
- ✗ B: 64° from miscomputing expression.
- ✗ D: 75° has no basis.

■ **TRAP ALERT:** Computing first angle and halving IT instead of its supplement.

■ **KEY INSIGHT:** (1) Compute angle. (2) Find supplement. (3) Halve the supplement.

■ **FAST METHOD:** $42^\circ \rightarrow$ supplement $138^\circ \rightarrow$ half 69° . ■ Desmos: evaluate $5x+3x^2$ at $x=3$.

Q20 — HARD

■ **Correct Answer: B) 60**

STEP-BY-STEP SOLUTION:

1. Co-interior sum to 180° : $(2a+b)+(a+2b) = 180$.
2. $3a+3b = 180 \rightarrow a+b = 60$.
3. The condition $a-b=6$ is extra info — not needed!

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: 54 needs sum= 162° .
- ✗ C: 66 needs sum= 198° .
- ✗ D: 72 needs sum= 216° .

■ **TRAP ALERT:** The extra condition $a-b=6$ distracts you into solving a system. The question only asks for $a+b$.

■ **KEY INSIGHT:** DSAT often gives more information than needed. Identify what is actually asked.

■ **FAST METHOD:** $3(a+b)=180 \rightarrow a+b=60$. No system needed!

Q21 — HARD

■ **Correct Answer: C) 135°**

STEP-BY-STEP SOLUTION:

1. $MR \perp PQ \rightarrow \text{angle } RMQ = 90^\circ$.
2. MS bisects RMQ $\rightarrow \text{angle } SMQ = 45^\circ$.
3. Angle PMS = $180^\circ - 45^\circ = 135^\circ$ (PMQ is a straight line).

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: 90° is angle PMR, not PMS.
- ✗ B: 120° has no basis.
- ✗ D: 150° would need $SMQ=30^\circ$.

■ **TRAP ALERT:** Stopping at 90° without adding the bisected portion.

■ **KEY INSIGHT:** Draw the figure: P—M—Q, MR up, MS between MR and MQ. PMS = $180^\circ - 45^\circ$.

■ **FAST METHOD:** $180^\circ - (90^\circ/2) = 135^\circ$.

Q22 — HARD

■ **Correct Answer: B) 35**

STEP-BY-STEP SOLUTION:

1. Three consecutive angles spanning one side = 180° .
2. $(2x+5)+(x+15)+(x+20) = 180 \rightarrow 4x+40 = 180 \rightarrow x = 35$
3. Check: $75^\circ+50^\circ+55^\circ = 180^\circ \checkmark$

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: $x=30 \rightarrow 65^\circ+45^\circ+50^\circ=160^\circ$.
- ✗ C: $x=40 \rightarrow 85^\circ+55^\circ+60^\circ=200^\circ$.
- ✗ D: $x=45 \rightarrow 95^\circ+60^\circ+65^\circ=220^\circ$.

■ **TRAP ALERT:** Using 360° instead of 180° for angles spanning a straight line.

■ **KEY INSIGHT:** Three consecutive angles spanning a straight line sum to 180° .

■ **FAST METHOD:** $4x+40=180 \rightarrow x=35$.

Q23 — HARD

■ **Correct Answer: C) 90°**

STEP-BY-STEP SOLUTION:

1. Co-interior sum to 180° : $(12k-6)+(6k+42) = 180$.
2. $18k+36 = 180 \rightarrow k = 8$.
3. Angle at line a = $12(8)-6 = 90^\circ$. Check: $6(8)+42 = 90^\circ$. Sum= $180^\circ \checkmark$

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: 78° needs $k=7$.
- ✗ B: 84° needs $k=7.5$.
- ✗ D: $96^\circ = 12(8)$, forgetting to subtract 6.

■ **TRAP ALERT:** Computing $12(8)=96$ and stopping without subtracting 6.

■ **KEY INSIGHT:** Always substitute fully: $12k-6$ means multiply THEN subtract.

■ **FAST METHOD:** $18k+36=180 \rightarrow k=8 \rightarrow 12(8)-6=90^\circ$.

Q24 — HARD

■ **Correct Answer: B) 45°**

STEP-BY-STEP SOLUTION:

1. Angle with horizontal = $\arctan(|\text{slope}|)$.
2. Slope of ■■ = 1 $\rightarrow$ angle = $\arctan(1) = 45^\circ$.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ A: 30° needs slope = $1/\sqrt{3}$.
- ✗ C: 60° needs slope = $\sqrt{3}$.
- ✗ D: 90° needs undefined slope (vertical).

■ **TRAP ALERT:** Thinking multiple lines through origin divide angles equally (60° each).

■ **KEY INSIGHT:** Angle with horizontal = $\arctan(|\text{slope}|)$. Slope=1 $\rightarrow 45^\circ$; $\sqrt{3} \rightarrow 60^\circ$; $1/\sqrt{3} \rightarrow 30^\circ$.

■ **FAST METHOD:** $\tan(\theta)=1 \rightarrow \theta=45^\circ$. ■ **Desmos:** plot $y=x$ and $y=0$.

Q25 — HARD

■ **Correct Answer: A) Correct: equal corresponding angles means the transversals are parallel.**

STEP-BY-STEP SOLUTION:

1. Both transversals make the same angle with the parallel lines.
2. Same inclination $\rightarrow$ same slope $\rightarrow$ the transversals are parallel.
3. This is the converse of the corresponding angles theorem.

WHY EACH WRONG ANSWER IS WRONG:

- ✗ B: Parallel transversals never meet, so "supplementary" is meaningless.
- ✗ C: The converse theorem applies symmetrically.
- ✗ D: $42^\circ + 42^\circ = 84^\circ \neq 90^\circ$, and the reasoning is invalid.

■ **TRAP ALERT:** Students doubt the claim because it seems too simple. The converse theorem is valid.

■ **KEY INSIGHT: CONVERSE:** If two lines make equal angles with a third line, those two lines are parallel.

■ **FAST METHOD:** Equal angles with parallel lines $\rightarrow$ same slope $\rightarrow$ parallel.

TOPIC SUMMARY — Lines & Angles

Common Mistakes (Top 3)

- Confusing ALTERNATE angles (equal) with CO-INTERIOR angles (supplementary = 180°). This accounts for ~40% of errors.
- Solving for the variable (x, k, m) and reporting THAT as the answer instead of substituting back for the ANGLE measure.
- Confusing COMPLEMENTARY (90°) with SUPPLEMENTARY (180°). Memory: C before S alphabetically, $90 < 180$.

Must Remember (Top 3 Rules)

- Vertical angles = EQUAL. Adjacent angles at intersection = SUPPLEMENTARY (180°).
- Parallel lines + transversal: Corresponding = Equal. Alternate Interior/Exterior = Equal. Co-interior = Supplementary (180°).
- Angles on one side of a straight line sum to 180° . All angles around a point sum to 360° .

Expected on DSAT

- Approximately 2–3 questions per test on lines and angles, in both modules.
- Most involve parallel lines and transversals with algebraic expressions.
- ■ Desmos: for questions with angle equations, graph to find intersection points quickly.

End of Solutions — Lines & Angles